Inductive Knowledge Graph Wave Networks

Giuseppe Pirrò
DeMaCS, University of Calabria
giuseppe.pirro@unical.it

Abstract

Inductive knowledge graph reasoning (IKGR) requires generalization to entities and relation types that have not been seen during training. We introduce Knowledge Graph Wave Networks (KGWN), an IKGR approach that learns adaptive per-relation wave operators to control the smoothing/sharpening of information propagation, along with velocity normalization ensuring stability at any propagation depth. Experiments demonstrate state-of-the-art results on inductive benchmarks and competitive transductive performance.

1 Introduction

Knowledge graphs (KGs) empower various applications [Hogan *et al.*, 2021]A central challenge is learning KG representations that generalize not only within a fixed graph (*transductive* setting), but also to entities and relations unseen during training (*inductive* setting) [Lee *et al.*, 2023]. Consider a movie KG where `Oppenheimer` is an *unseen entity* at training time. To answer the link prediction query (`Oppenheimer`, `directed_by`, ?), a model cannot rely on memorized entity embeddings; instead, it must infer the missing director by propagating evidence through relation patterns (e.g., shared collaborators, studios). Likewise, for (`Oppenheimer`, `genre`, ?), evidence can flow through thematic connections (e.g., nearby films), enabling prediction of genres. Effective inductive KG reasoning (IKGR) thus demands *relation-dependent propagation*: selectively gathering evidence through heterogeneous, directed relations while preserving distinct semantic signals. This is challenging under *relational heterophily* [Qian *et al.*, 2025] where homophilic relations like `same_genre` connect same-type entities and benefit from smoothing, while heterophilic relations like `directed_by` connect different entity types and require sharpening to preserve type distinctions.

Related Work. GraIL [Teru *et al.*, 2020], NBFNet [Zhu *et al.*, 2021], SNRI [Xu *et al.*, 2022], MulGA [Shang *et al.*, 2024], and RED-GNN [Zhang and Yao, 2022] reason over query-conditioned local subgraphs to accommodate unseen entities only. IKGR-ME [Cui *et al.*, 2022] extends entity induction to multi-batch emerging entities. InGram [Lee *et al.*, 2023] supports both new entities and relations at test time. All these methods rely on discrete, neighbor-centric aggregation that inherently biases toward low-pass smoothing, lacking an explicit mechanism to control when smoothing (low-pass) versus sharpening (high-pass) should be applied. As depth increases, this can lead to oversmoothing that washes out relational cues crucial for inductive inference [Qian *et al.*, 2025; He *et al.*, 2025]—particularly problematic for heterophilic relations requiring sharp cross-type distinctions. Diffusion models like DiffusionE [Cao *et al.*, 2024] treat propagation as continuous-time heat flow, progressively smoothing node signals; however, diffusion behaves primarily as a low-pass filter, blurring signals in heterophilic relations. GWN [Yue *et al.*, 2025] and EWGNN [Wu *et al.*, 2025] offer a conservative alternative that maintains high-frequency signals, but both are limited to *homogeneous, transductive* settings.

Table 1: Comparison of KGWN to relevant approaches.

Method	KG	Entity Induction	Relation Induction	Heterophily-aware Propagation	Stability Guarantees
GraIL (ICML'20)	✓	✗	✗	✗	✗
NBFNet (NeurIPS'21)	✓	✓	✗	✗	✗
SNRI (IJCAI'22)	✓	✓	✗	✗	✗
RED-GNN (WebConf '22)	✓	✓	✗	✗	✗
IKGR-ME (CIKM'22)	✓	✓	✗	✗	✗
InGram (ICML'23)	✓	✓	✓	✗	✗
MulGA (IJCAI'24)	✓	✓	✗	✗	✗
GRAND (NeurIPS'21)	✗	✗	✗	✗	✓
DiffusionE (KDD'24)	✓	✓	✗	✗	✗
GWN (WebConf'25)	✗	✗	✗	✗	✓
EWGNN (IJCAI'25)	✗	✗	✗	✗	✓
KGWN (Ours)	✓	✓	✓	✓	✓

Our approach. Table 1 shows that existing approaches fail to address stable relation-dependent propagation in a full IKGR scenario. Thus, we introduce Knowledge Graph Wave Networks (KGWN) defining relation-specific propagation channels with each relation type inducing a learnable *wave operator* with two key parameters: (i) *propagation speeds* controlling signal strength and reach, and (ii) *frequency gates* enabling homophilic relations to smooth while heterophilic relations to sharpen, preserving cross-type distinctions. Since mode parameters are learned per-relation, they transfer directly to unseen entities. For unseen relations, KGWN computes relation representations leveraging a *relation graph* representation.

Contributions and outline. We contribute: **(1)** a multi-relational wave equation for directed KGs using symmetric operators (§3.1) and velocity normalization ensuring bounded propagation at any depth with uniform stability guarantees (§3.2); **(2)** relation-adaptive frequency control explicitly addressing relational heterophily (§3.3); **(3)** the first wave-based framework for inductive KG reasoning that generalizes to unseen entities *and* relations (§4).

2 Preliminaries

A knowledge graph is a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, where $\mathcal{V}$ contains N entities, $\mathcal{R}$ is a set of relation types, and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$ directed triples.

2.1 Knowledge Graph Reasoning Settings

We focus on *link prediction*, which operates under two settings (i) *transductive*: all entities and relations at test time were present during training, and the model predicts missing edges among known elements; (ii) *inductive*: the model needs to generalize to new entities/relations without retraining. We partition $\mathcal{V} = \mathcal{V}_{\text{train}} \sqcup \mathcal{V}_{\text{test}}$ and $\mathcal{R} = \mathcal{R}_{\text{train}} \sqcup \mathcal{R}_{\text{test}}$. Induction has three levels: (1) *entity-level*—unseen entities, known relations; (2) *relation-level*—unseen relations, known entities; (3) *fully inductive*—both novel entities and relations, with at least one known component per query.

Embedding initializations. We initialize entity and relation embeddings using pretrained text encoders [Sanh *et al.*, 2019] applied to their textual descriptions:

$$\mathbf{X}^{(0)} = \text{TextEnc}\big(\{\text{name}_v, \text{desc}_v\}_{v\in\mathcal{V}}\big),$$
$$\mathbf{R}^{(0)} = \text{TextEnc}\big(\{\text{name}_r, \text{desc}_r\}_{r\in\mathcal{R}}\big).$$

3 Knowledge Graph Wave Networks

We now introduce *Knowledge Graph Wave Networks* (KGWN) for fully inductive knowledge graph reasoning. Fig. 1 gives an overview. After initial embeddings, KGWN defines symmetric wave operators with controlled spectrum to enable multi-relational wave propagation (§3.1), then multirelational aggregation with stability guarantees (§3.2), and finally introduce relation-specific frequency adaptation via *relation gates*, allowing the model to learn whether relation should act as a smoothing or sharpening operator (§3.3).

3.1 Symmetric Multi-Relational Operators

The standard wave equation $\ddot{\mathbf{X}}(t) = \mathbf{L}\mathbf{X}(t)$ propagates signals using a *single symmetric* graph operator $\mathbf{L}$, yielding wave-like dynamics that can preserve high-frequency components and mitigate oversmoothing on homogeneous graphs [Yue *et al.*, 2025]. In contrast, knowledge graphs are *multi-relational* and *directed*, with edges typed by relations $r \in \mathcal{R}$. To retain the stability and spectral control afforded by wave dynamics, KGWN assigns each relation its own *symmetric* propagation operator $\mathbf{L}_r$, and combines them into a controlled multi-relational operator for propagation. To handle directionality, we define two KGWN variants:

1. KGWN-S (Symmetric Channel). For each relation $r \in \mathcal{R}$, $\tilde{\mathbf{A}}_r := \mathbf{A}_r + \mathbf{A}_r^\top$ is its symmetrized adjacency, where $\mathbf{A}_r$ is the directed adjacency matrix containing edges of type r. Let $\mathbf{D}_r = \text{diag}(\tilde{\mathbf{A}}_r\mathbf{1})$ denote the corresponding degree matrix. We construct the normalized operator

$$\mathbf{P}_r = \mathbf{D}_r^{-1/2}\,\tilde{\mathbf{A}}_r\,\mathbf{D}_r^{-1/2}, \qquad \mathbf{L}_r := \mathbf{P}_r - \mathbf{I}. \tag{1}$$

Symmetrization treats each triple (u, r, v) as bidirectional, pooling information from both head-to-tail and tail-to-head propagation. Each $\mathbf{L}_r$ is symmetric with spectrum $\text{spec}(\mathbf{L}_r) \subseteq [-2, 0]$: symmetrization yields a symmetric normalized adjacency $\mathbf{P}_r$ with $\text{spec}(\mathbf{P}_r) \subseteq [-1, 1]$, and the shift $\mathbf{L}_r = \mathbf{P}_r - \mathbf{I}$ moves the spectrum to $[-2, 0]$ (Lemma 1, Appendix A). This makes the continuous-time dynamics $\ddot{\mathbf{X}} = \mathbf{L}\mathbf{X}$ oscillatory (wave-like) rather than exponentially growing, and provides a spectral bound used for stability (Eq. (9)).

2. KGWN-L: (Lifted Channel). Symmetrization discards directional information, which may be semantically important (e.g., `parent_of` vs. `child_of`). To preserve direction while maintaining symmetry, we use a lifted construction. For each relation r, we construct the adjacency

$$\mathbf{B}_r := \begin{bmatrix} \mathbf{0} & \mathbf{A}_r \\ \mathbf{A}_r^\top & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{2N\times 2N}, \tag{2}$$

which encodes directed edges via block structure: the upper-right block contains forward edges, the lower-left block contains reverse edges. Let $\mathbf{D}_r^{(\text{lift})} = \text{diag}(\mathbf{B}_r\mathbf{1})$ be the degree matrix for this lifted graph. We then define:

$$\mathbf{P}_r^{(\text{lift})} = (\mathbf{D}_r^{(\text{lift})})^{-1/2}\,\mathbf{B}_r\,(\mathbf{D}_r^{(\text{lift})})^{-1/2}, \;\; \mathbf{L}_r^{(\text{lift})} := \mathbf{P}_r^{(\text{lift})} - \mathbf{I}. \tag{3}$$

Consider a simple example with two entities $\{E_1, E_2\}$ and one directed edge (E_1, r, E_2) (e.g., $r =$ `parent_of`). The lifted graph duplicates each entity into forward and backward copies: $\{E_{1,f}, E_{1,b}, E_{2,f}, E_{2,b}\}$. The block matrix $\mathbf{B}_r$ encodes directed edges via its blocks: (i) in the upper-right block, $E_{1,f} \to E_{2,b}$ (original direction); (ii) in the lower-left block, $E_{2,f} \to E_{1,b}$ (reverse direction), induced by $\mathbf{A}_r^\top$. After propagation on this $2N$-node bipartite graph (forward vs. backward partition), the final embedding for entity E_i fuses the two context-specific states $E_{i,f}$ (as head) and $E_{i,b}$ (as tail) via a fusion layer. Despite encoding directed edges, $\mathbf{B}_r$ is symmetric by construction, so $\mathbf{L}_r^{(\text{lift})}$ inherits the same spectral properties as the symmetric variant: spectrum in $[-2, 0]$, enabling stable wave propagation. The lifted construction operates on state vectors $\mathbf{X}^{(n)} \in \mathbb{R}^{2N\times d}$ that maintain separate representations for each entity in forward and backward contexts (see Lemma 2 in Appendix A).

In what follows, we use $\mathbf{L}_r$ to denote the relation operator under either construction. We next show how to combine per-relation operators into a unified wave equation.

3.2 Multi-Relational Aggregation and Stability

To combine the per-relation operators $\{\mathbf{L}_r\}_{r\in\mathcal{R}}$ into a single multi-relational wave equation, a natural choice would be a weighted sum $\mathbf{L}_{\text{agg}} = \sum_{r\in\mathcal{R}} w_r\,\mathbf{L}_r$. The key issue here is stability: even if each $\mathbf{L}_r$ is individually well-behaved, an unconstrained choice of weights can increase the spectral radius of $\mathbf{L}_{\text{agg}}$, which in turn can make the update in Eq. (7)

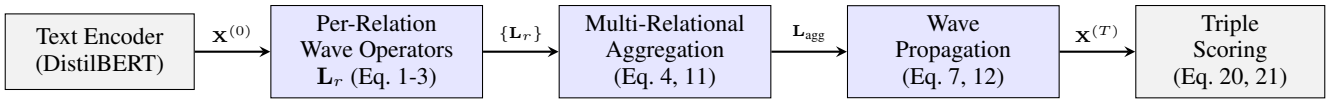


Figure 1: KGWN pipeline overview.

unstable. We therefore need a condition (and a normalization) that controls $\text{spec}(\mathbf{L}_{\text{agg}})$. Let $\{a_r > 0\}_{r\in\mathcal{R}}$ be learnable relation velocities controlling each relation's propagation strength. We define the aggregated operator

$$\mathbf{L}_{\text{agg}} = \sum_{r\in\mathcal{R}} a_r^2\, \mathbf{L}_r, \tag{4}$$

yielding the multi-relational wave equation

$$\ddot{\mathbf{X}}(t) = \mathbf{L}_{\text{agg}}\, \mathbf{X}(t). \tag{5}$$

The squared velocities a_r^2 ensure non-negativity, which is essential for spectral control: since each $\mathbf{L}_r$ has spectrum in $[-2, 0]$ and velocities are nonnegative, using Rayleigh quotient bounds (Theorem 1 in Appendix A), we obtain

$$\text{spec}(\mathbf{L}_{\text{agg}}) \subseteq [-2\nu^2, 0], \quad \text{where} \quad \nu^2 = \sum_{r\in\mathcal{R}} a_r^2. \tag{6}$$

To ensure boundedness, we require $|2 + \tau^2\lambda| \leq 2$ for all $\lambda \in \text{spec}(\mathbf{L}_{\text{agg}})$. Since $\text{spec}(\mathbf{L}_{\text{agg}}) \subseteq [-2\nu^2, 0]$ from Eq. (6), the worst case occurs at the most negative eigenvalue $\lambda = -2\nu^2$. Substituting this into the stability condition gives $|2 - 2\tau^2\nu^2| \leq 2$, which simplifies to $\tau^2\nu^2 \leq 2$. We use the strict inequality $\tau^2\nu^2 < 2-\delta$ with safety margin $\delta > 0$ to account for numerical precision and ensure a buffer for learning dynamics. This leads directly to the velocity normalization scheme in Eq. (10). To implement wave propagation computationally, we discretize the continuous eq. (5) in time. The second-order wave equation $\ddot{\mathbf{X}}(t) = \mathbf{L}_{\text{agg}}\mathbf{X}(t)$ requires a numerical integrator that preserves good long-time energy behavior. We employ the *leapfrog* scheme [Hairer *et al.*, 2006]. The idea is to approximate the second derivative $\ddot{\mathbf{X}}$ via a centered finite difference, leading to the discrete update

$$\mathbf{X}^{(n+1)} = \left(2\mathbf{I} + \tau^2\mathbf{L}_{\text{agg}}\right)\mathbf{X}^{(n)} - \mathbf{X}^{(n-1)} \tag{7}$$

where $\tau > 0$ is the time step and $\mathbf{X}^{(n)} \approx \mathbf{X}(n\tau)$ denotes the state at discrete time $n\tau$.

Stability note. The leapfrog update in Eq. (7) is second-order accurate in τ and has good long-time energy behavior, but it is stable only if every eigenmode stays bounded. Let λ be an eigenvalue of $\mathbf{L}_{\text{agg}}$ and consider the corresponding scalar mode $x^{(n)}$. Substituting $x^{(n)}$=ρ^n into Eq. (7) gives the characteristic equation $\rho^2 - (2 + \tau^2\lambda)\rho + 1 = 0$. Boundedness (i.e., $|\rho| \leq 1$ for both roots) is equivalent to the standard condition (Theorem 2 in Appendix B):

$$|2 + \tau^2\lambda| \leq 2 \qquad \text{for all } \lambda \in \text{spec}(\mathbf{L}_{\text{agg}}), \tag{8}$$

Using the spectral bound $\text{spec}(\mathbf{L}_{\text{agg}}) \subseteq [-2\nu^2, 0]$ from Eq. (6), the worst case is the most negative eigenvalue λ=$-2\nu^2$. Plugging this into Eq. (8) yields

$$\tau^2\nu^2 < 2, \tag{9}$$

which hints that the time step τ must be small enough to resolve the fastest oscillations (largest $|\lambda|$). Rather than hand-tuning τ or manually constraining the learned per-relation velocities $\{a_r\}$, we enforce Eq. (9) via a global rescaling:

$$\tilde{a}_r = \frac{a_r}{\max\left(1,\ \frac{\tau\nu}{\sqrt{2-\delta}}\right)}, \tag{10}$$

where $\delta \in (0, 1)$ is a small safety margin and $\nu = \sqrt{\sum_{r\in\mathcal{R}} a_r^2}$. Let $\tilde{\nu}^2 = \sum_{r\in\mathcal{R}} \tilde{a}_r^2$ denote the post-rescaling aggregate velocity. By construction, $\tau^2\tilde{\nu}^2 \leq 2 - \delta$, so every eigenmode remains bounded for any propagation depth. We denote the post-normalization velocities by $\tilde{a}_r$ for clarity, and use these rescaled values in all subsequent propagation steps. For notational simplicity in later equations, we write a_r with the understanding that it refers to the normalized quantity $\tilde{a}_r$ from Eq. (10). See Appendix B for the detailed derivations.

While stability prevents numerical blow-up, it does not address a key KG challenge: relations are heterogeneous and may require different frequency responses (e.g., smoothing for homophilic relations vs. sharpening for heterophilic ones).

3.3 Relation-Level Frequency Adaptation

The aggregated operator in Eq. (4) weights relations by strength via the learned velocities $\{a_r\}$, but it does not allow different relations to exhibit different frequency behaviors (smoothing vs. sharpening). Hence, we introduce *relation gates* that separate (i) *importance* and (ii) *frequency behavior*. Let $\beta_r \geq 0$ with $\sum_{r\in\mathcal{R}} \beta_r = 1$ be learned importance weights (e.g., $\beta_r = \text{softmax}(\gamma_r)$), and let $\alpha_r \in [-1, 1]$ be signed frequency gates (positive = smoothing, negative = sharpening). We retain the learned velocities $a_r > 0$ from Eq. (4) to control coupling strength. The adaptive aggregated operator now becomes

$$\tilde{\mathbf{L}}_{\text{agg}} := \sum_{r\in\mathcal{R}} \beta_r\, \alpha_r\, a_r^2\, \mathbf{L}_r. \tag{11}$$

Allowing $\alpha_r < 0$ can flip the sign of some terms and introduce positive eigenvalues (Lemma 3 in Appendix C). When $\alpha_r < 0$ (sharpening), $\tilde{\mathbf{L}}_{\text{agg}}$ may acquire positive eigenvalues, which induce exponential growth in the corresponding eigenmodes. Constraining $\alpha_r \geq 0$ would eliminate this instability but also remove the model's ability to sharpen signals along heterophilic relations. We stabilize positive-eigenvalue modes via an anchoring term toward the initialization. This acts as a restoring force that becomes stronger as representations drift further from $\mathbf{X}^{(0)}$, preserving both stability and expressive power. The anchored update is:

$$\mathbf{X}^{(n+1)} = \left(2\mathbf{I} + \tau^2\widehat{\mathbf{L}}_{\text{agg}}\right)\mathbf{X}^{(n)} - \mathbf{X}^{(n-1)} + \varepsilon\left(\mathbf{X}^{(0)} - \mathbf{X}^{(n)}\right) \tag{12}$$

where $\varepsilon \in [0, 1)$ controls the anchor strength. We note that β_r controls *how much* relation r contributes overall, while α_r

controls *how* it acts spectrally: $\alpha_r > 0$ encourages smoothing along homophilic relations, whereas $\alpha_r < 0$ encourages sharpening across heterophilic ones. Moreover, note that if we restrict $\alpha_r \in [0, 1]$, then $\tilde{\mathbf{L}}_{\text{agg}}$ remains negative semidefinite, and leapfrog stability follows the condition in §3.2. When allowing signed gates $\alpha_r \in [-1, 1]$, $\tilde{\mathbf{L}}_{\text{agg}}$ may have positive eigenvalues; the anchor in Eq. (12) stabilizes these modes (Theorem 3 in Appendix C). A sufficient condition is:

$$\tau^2 \|\widehat{\mathbf{L}}_{\text{agg}}\|_2 \leq \varepsilon, \tag{13}$$

We enforce Eq. (13) by rescaling the adaptive operator. Let $\kappa := \sum_{r\in\mathcal{R}} \beta_r \, |\alpha_r| \, a_r^2$ and recall $\|\mathbf{L}_r\|_2 \leq 2$ for the symmetric construction (Appendix C.3). Then $\|\tilde{\mathbf{L}}_{\text{agg}}\|_2 \leq 2\kappa$ and it suffices to ensure $2\tau^2\kappa \leq \varepsilon$. We therefore set

$$\widehat{\mathbf{L}}_{\text{agg}} := \eta \, \tilde{\mathbf{L}}_{\text{agg}}, \qquad \eta = \min\left(1, \frac{\varepsilon - \delta}{2\tau^2\kappa}\right), \tag{14}$$

with margin $\delta \in (0, \varepsilon)$, and use $\widehat{\mathbf{L}}_{\text{agg}}$ in Eq. (12).

4 Inductive Reasoning

KGWN can work in both entity and full inductive scenarios.

4.1 Entity Induction

Consider a link prediction query $(h, r, ?)$ (or $(?, r, t)$), where h (or t) may be unseen during training. Rather than relying on entity-specific embeddings, KGWN performs reasoning over the *test-time* graph structure. For a candidate tail t, KGWN extracts the k-hop enclosing subgraph around the pair (h, t):

$$\mathcal{V}_{h,t}^{(k)} \;=\; \mathcal{N}_k(h) \cup \mathcal{N}_k(t),$$

where $\mathcal{N}_k(v)$ denotes the set of nodes within k hops of v in the test graph $\mathcal{G}_{\text{test}}$ (computed on the undirected view for extraction). The induced subgraph $\mathcal{G}_{h,t}^{(k)} = (\mathcal{V}_{h,t}^{(k)}, \mathcal{E}_{h,t}^{(k)}, \mathcal{R})$ is then the domain for wave propagation, with

$$\mathcal{E}_{h,t}^{(k)} \;=\; \{(u, r', v) \in \mathcal{E}_{\text{test}} \,:\, u, v \in \mathcal{V}_{h,t}^{(k)}\}.$$

The crucial observation is that KGWN's dynamics depend only on relation types, not entity identities. On the extracted subgraph, we construct per-relation operators $\mathbf{L}_r^{(k;h,t)}$ following the same recipe as Eqs. (1)–(3), then aggregate:

$$\mathbf{L}_{\text{agg}}^{(k;h,t)} = \sum_{r\in\mathcal{R}} a_r^2 \, \mathbf{L}_r^{(k;h,t)}. \tag{15}$$

The relation parameters $\{a_r\}$ (and optionally $\{\alpha_r, \beta_r\}$) come directly from training—no fine-tuning required. Wave propagation proceeds via the standard leapfrog update:

$$\mathbf{X}^{(n+1)} = \big(2\mathbf{I} + \tau^2 \mathbf{L}_{\text{agg}}^{(k;h,t)}\big)\mathbf{X}^{(n)} - \mathbf{X}^{(n-1)}, \tag{16}$$

with initial states obtained from the text encoder. Since text embeddings are available for any entity with a name or description, unseen elements pose no obstacle.

4.2 Relation and Full Induction

When test queries involve unseen relation types, entity induction alone does not suffice. KGWN addresses this by propagating wave dynamics over a *relation graph* $\mathcal{G}_{\text{rel}} = (\mathcal{R}, \mathcal{E}_{\text{rel}})$, where edges connect relations that co-occur in training triples (see Appendix D.4). We initialize relation embeddings $\mathbf{R}^{(0)} \in \mathbb{R}^{|\mathcal{R}|\times d_r}$ from text and propagate via the same leapfrog scheme with normalized Laplacian $\mathbf{L}_{\text{rel}}$:

$$\mathbf{R}^{(m+1)} = \big(2\mathbf{I} + \tau_{\text{rel}}^2 \mathbf{L}_{\text{rel}}\big)\mathbf{R}^{(m)} - \mathbf{R}^{(m-1)}. \tag{17}$$

After M propagation steps, learnable projections $(\mathbf{w}_\cdot, b_\cdot)$ decode each embedding $\mathbf{r}_r^{(M)}$ into the wave parameters (§3.3):

$$\begin{aligned} a_r &= \text{softplus}(\mathbf{w}_a^\top \mathbf{r}_r^{(M)} + b_a) \\ \alpha_r &= \tanh(\mathbf{w}_\alpha^\top \mathbf{r}_r^{(M)} + b_\alpha) \\ \beta_r &= \text{softmax}_r(\mathbf{w}_\beta^\top \mathbf{r}_r^{(M)} + b_\beta). \end{aligned} \tag{18}$$

The activations enforce the constraints in §3.3: softplus ensures $a_r > 0$ (positive wave velocities), $\tanh$ bounds $\alpha_r \in (-1, 1)$ (enabling smoothing/sharpening), and softmax normalizes the mixing weights $\{\beta_r\}$ across relations. For unseen relations r_{new}, we insert their text embeddings into $\mathcal{G}_{\text{rel}}$, connect them to similar known relations, rerun Eq. (17), and decode parameters via Eq. (18) without gradient updates. These inferred parameters drive propagation in the query subgraph:

$$\tilde{\mathbf{L}}_{\text{agg}}^{(k;h,t)} = \sum_{r\in\mathcal{R}_{h,t}} \tilde{\beta}_r \, \alpha_r \, a_r^2 \, \mathbf{L}_r^{(k;h,t)} \tag{19}$$

where $\mathcal{R}_{h,t}$ denotes relations in the subgraph and $\tilde{\beta}_r$ renormalizes the mixing weights. Entity embeddings then evolve via the adaptive update (Eq. 12).

Scoring. For *link prediction* in the entity induction setting, we use MLP scoring to match DiffusionE [Cao *et al.*, 2024]:

$$s(h, r, t) = \text{MLP}\Big(\big[\mathbf{x}_h^{(T)} \,\|\, \mathbf{x}_t^{(T)} \,\|\, (\mathbf{x}_h^{(T)} \odot \mathbf{x}_t^{(T)}) \,\|\, \mathbf{r}_r^{(M)}\big]\Big) \tag{20}$$

For the full inductive setting, we adopt the bilinear scoring function to match InGram [Lee *et al.*, 2023]:

$$s(h, r, t) = \mathbf{x}_h^{(T)\top} \text{diag}(\mathbf{W}\mathbf{r}_r^{(M)}) \, \mathbf{x}_t^{(T)}, \tag{21}$$

where the diagonal matrix weights each embedding dimension according to the relation, enabling efficient yet expressive head–tail matching.

4.3 Training

For *entity induction*, each positive triple $(h, r, t) \in \mathcal{T}^+$ is paired with K corrupted triples $(\mathcal{T}^-)$ obtained by replacing the head or tail. We minimize the binary cross-entropy:

$$\mathcal{L}_{\text{ent}} = -\sum_{(h,r,t)\in\mathcal{T}^+} \Bigg[\log\sigma\big(s(h, r, t)\big) + \frac{1}{K}\sum_{(h',r,t')\in\mathcal{T}^-} \log\big(1 - \sigma(s(h', r, t'))\big)\Bigg] \tag{22}$$

In the *full inductive* setting, following InGram [Lee *et al.*, 2023], we train with margin ($\gamma > 0$) to rank candidates:

$$\mathcal{L}_{\text{full}} = \sum_{(h,r,t)\in\mathcal{T}^+} \frac{1}{K} \sum_{(h',r,t')\in\mathcal{T}^-} \max\big(0,\; \gamma - s(h, r, t) + s(h', r, t')\big) \tag{23}$$

5 Experimental Evaluation

We evaluate KGWN across inductive/transductive settings:

- **Main Results:** Entity induction (RQ1), relation induction (RQ2), and transductive reasoning (RQ3)
- **Analysis:** Component ablations and hyperparameter sensitivity (RQ4).

Evaluation protocol. We evaluate link prediction in the filtered setting [Bordes *et al.*, 2013]. For each test triple (h, r, t), we rank tail candidates (h, r, t') and head candidates (h', r, t) for all $t', h' \in \mathcal{V}_{\text{test}}$. For each candidate, we extract its k-hop enclosing subgraph (default $k = 2$), propagate with KGWN, and compute the score. We train KGWN using Adam (lr 5×10^{-4}), batch size 128, 50 epochs with early stopping (patience 3) on validation MRR. Default hyperparameters (d=128, k=2, T=3, τ=0.05, ε=0.2, δ=0.15) are identical across all benchmarks without per-dataset tuning (full table in Appendix E); sensitivity is analysed in §5.4. Text features use DistilBERT [Sanh *et al.*, 2019]. Implementation details including Algorithm 1 (pseudocode) and complexity analysis are in Appendix D. We report ranking metrics Hits@1/10 and MRR. Code is available online*.

The two settings use different datasets because they follow distinct community protocols: entity splits (V1–V4) were introduced by [Teru *et al.*, 2020]; NELL/Wikidata68K/Freebase overlap splits were introduced by [Lee *et al.*, 2023] as the only available benchmarks with disjoint unseen-relation test sets, which the entity-induction datasets do not provide.

5.1 Entity-Level Induction (RQ1)

We used the FB15k-237, WN18RR, and NELL-995 benchmarks and splits by [Teru *et al.*, 2020]. We compare against recent KG-based methods evaluated on the same splits: GraIL [Teru *et al.*, 2020], SNRI [Xu *et al.*, 2022], NBFNet [Zhu *et al.*, 2021], RED-GNN [Zhang and Yao, 2022] (R-GNN), MulGA [Shang *et al.*, 2025], and DiffusionE [Cao *et al.*, 2024] (DiffE). For conciseness, we omit baselines that underperformed based on the recent evaluation by [Cao *et al.*, 2024]. Table 2 reports entity-inductive link prediction across four increasingly difficult splits per benchmark. KGWN-L achieves state-of-the-art on all configurations, while KGWN-S consistently outperforms DiffusionE. DiffusionE's first-order diffusion inherently smooths signals, losing discriminative power on heterophilic relations, whereas KGWN's second-order wave equation preserves high-frequency structure through oscillatory propagation. On WN18RR-V3, KGWN-L improves MRR by $+3.1$ pp over DiffusionE (.491 vs .460), showing that taxonomic relations benefit from bidirectional wave flow. KGWN-L's learnable per-relation parameters (a_r, α_r) enable adaptive smoothing/sharpening, yielding MRR improvement over DiffusionE on V1 (.502 vs .310) and consistently outperforming MulGA across all splits. NELL-995 is extracted via noisy web extraction. KGWN-L achieves .831 MRR on V1, surpassing MulGA (.812) by leveraging text-initialized features that provide semantic grounding. The gap widens on harder splits ($+0.018$ points on both V3 and V4), where wave dynamics better separate signal from noise and KGWN scales gracefully while SNRI runs out-of-memory. KGWN-S offers a favorable accuracy-efficiency trade-off, matching or exceeding DiffusionE. Unlike DiffusionE's learned semantic modules, KGWN's wave operators are derived directly from adjacency structure, reducing learnable parameters to a few scalars per relation. On the hardest V3/V4 splits with sparser test subgraphs, KGWN-L improves over the next-best method by $+1.5$ pp and $+2.3$ pp, respectively.

*https://github.com/giuseppepirro/KGWN

Table 2: Entity-inductive link prediction. V1-V4 denote increasingly difficult test splits with decreasing entity overlap with training. Results for competitors are from *Cao et al., 2024*; MulGA from *Shang et al., 2024*. Best results in **bold**. KGWN results are averaged over 3 runs with 3 seeds; standard deviations and significance tests are in Appendix E.7.

Method	WN18RR				FB15k-237				NELL-995			
	V1	V2	V3	V4	V1	V2	V3	V4	V1	V2	V3	V4
MRR												
GraIL	.627	.625	.323	.553	.279	.276	.251	.227	.481	.297	.322	.262
SNRI	.576	.539	.305	.486	.186	.227	OM	OM	.363	.200	OM	OM
NBFNet	.741	.704	.452	.641	.422	.514	.476	.453	.584	.410	.425	.287
R-GNN	.701	.690	.427	.651	.369	.469	.445	.442	.637	.419	.436	.363
MulGA	.747	.705	.476	.655	.485	.538	.492	.472	.812	.525	.510	.429
DiffE	.727	.724	.460	.660	.310	.474	.456	.446	.673	.424	.458	.308
KGWN-S	.738	.726	.469	.668	.328	.489	.468	.458	.702	.448	.475	.332
KGWN-L	**.761**	**.732**	**.491**	**.678**	**.502**	**.556**	**.511**	**.489**	**.831**	**.542**	**.528**	**.447**
Hits@1												
GraIL	55.4	54.2	27.8	44.3	20.5	20.2	16.5	14.3	42.5	19.9	22.4	15.3
SNRI	47.9	44.8	23.4	39.9	10.7	14.4	OM	OM	18.5	12.8	OM	OM
NBFNet	69.5	65.1	39.2	60.8	33.5	42.1	38.4	36.0	50.0	27.1	26.2	23.3
R-GNN	65.3	63.3	36.8	60.6	30.2	38.1	35.1	34.0	52.5	31.9	34.5	25.9
MulGA	70.8	65.2	42.4	61.1	40.2	44.4	39.2	37.2	75.0	42.5	42.2	34.1
DiffE	68.5	67.8	41.0	61.5	22.8	38.2	36.8	35.4	59.2	32.8	36.5	24.6
KGWN-S	69.8	68.2	41.8	62.4	24.5	39.6	37.8	36.2	62.5	35.2	38.4	26.8
KGWN-L	**72.4**	**68.6**	**44.8**	**64.2**	**42.5**	**46.8**	**42.1**	**39.5**	**77.3**	**44.8**	**44.6**	**36.5**
Hits@10												
GraIL	76.0	77.6	40.9	68.7	42.9	42.4	42.4	38.9	56.5	49.6	51.8	50.6
SNRI	75.0	69.8	43.2	65.2	33.2	38.8	OM	OM	55.5	33.3	OM	OM
NBFNet	82.6	79.8	56.8	69.4	57.4	68.5	63.7	62.7	79.5	63.5	60.6	59.1
R-GNN	79.9	78.0	52.4	72.1	48.3	62.9	60.3	62.1	86.6	60.1	59.4	55.6
MulGA	82.7	80.1	56.0	73.6	62.2	70.7	66.2	65.2	93.5	70.9	67.9	57.1
DiffE	85.7	84.3	60.5	73.7	52.5	66.1	62.7	62.5	87.3	66.0	62.0	52.3
KGWN-S	85.9	84.5	61.2	74.5	54.2	67.8	64.2	63.8	89.5	68.2	64.5	54.8
KGWN-L	**86.4**	**85.1**	**62.3**	**76.8**	**64.5**	**72.4**	**68.9**	**67.8**	**95.2**	**73.2**	**70.5**	**59.8**

5.2 Relation-Level Induction (RQ2)

We evaluate now *both* unseen relations and entities at test time. We use the benchmarks from InGram [Lee *et al.*, 2023] spanning NELL [Xiong *et al.*, 2017], Wikidata68K [Gesese *et al.*, 2022], and Freebase [Bollacker *et al.*, 2008], with varying overlap (100%, 75%, 50%, 25%). We compare against GraIL [Teru *et al.*, 2020], NBFNet [Zhu *et al.*, 2021], RED-GNN [Zhang and Yao, 2022], and InGram [Lee *et al.*, 2023]. Table 3 shows that KGWN-L achieves state-of-the-art across all configurations. InGram pioneered relation-level induction via relation graph construction; KGWN builds on this foundation but replaces static message passing with wave-based dynamics. On NELL-25 (only 25% relations seen), KGWN-L improves MRR by $+11.1\%$ over InGram (.371 vs .334), showing that oscillatory propagation captures higher-order relational interactions missed by static aggregation. As

Table 3: Relation-inductive results with different % relations shared with training. Results for the competitors are taken from *Lee at al., 2023*. KGWN results are averaged over 3 runs; see Appendix E.7 for variance and significance analysis.

Method	NELL				Wikidata				Freebase			
	100	75	50	25	100	75	50	25	100	75	50	25
MRR												
GraIL	.135	.096	.162	.216	–	–	–	–	–	–	–	–
NBFNet	.096	.137	.225	.283	.014	.072	.062	.154	.072	.089	.130	.224
R-GNN	.212	.203	.179	.214	.096	.172	.058	.170	.121	.107	.129	.145
InGram	.309	.261	.281	.334	.107	.247	.068	.186	.223	.189	.117	.133
KGWN-S	.318	.268	.289	.345	.114	.255	.075	.193	.230	.196	.126	.141
KGWN-L	**.342**	**.291**	**.312**	**.371**	**.128**	**.278**	**.091**	**.215**	**.251**	**.218**	**.148**	**.168**
Hits@1												
GraIL	.114	.036	.104	.160	–	–	–	–	–	–	–	–
NBFNet	.032	.077	.161	.224	.005	.028	.036	.092	.026	.048	.071	.137
R-GNN	.114	.129	.115	.166	.070	.110	.033	.111	.053	.057	.072	.077
InGram	.212	.167	.193	.241	.072	.179	.034	.124	.146	.119	.067	.067
KGWN-S	.219	.174	.200	.250	.079	.186	.041	.131	.153	.126	.075	.076
KGWN-L	**.241**	**.196**	**.223**	**.278**	**.095**	**.208**	**.058**	**.154**	**.176**	**.148**	**.096**	**.101**
Hits@10												
GraIL	.173	.205	.288	.366	–	–	–	–	–	–	–	–
NBFNet	.199	.255	.346	.417	.026	.172	.105	.301	.154	.166	.259	.410
R-GNN	.385	.353	.280	.266	.136	.290	.093	.263	.263	.201	.251	.284
InGram	.506	.464	.453	.501	.169	.362	.135	.309	.371	.325	.218	.271
KGWN-S	.517	.475	.464	.514	.179	.373	.145	.320	.382	.337	.230	.283
KGWN-L	**.548**	**.506**	**.495**	**.547**	**.201**	**.398**	**.168**	**.348**	**.412**	**.368**	**.261**	**.315**

known relations decrease (100→25), most methods collapse: NBFNet's MRR on Freebase jumps erratically (.072→.224), revealing brittleness. KGWN-L degrades gracefully, maintaining consistent improvements over InGram ($+11.1\%$ on NELL, $+15.6\%$ on Wikidata, $+26.3\%$ on Freebase at 25-split). On heterogeneous Wikidata and Freebase, where GraIL fails entirely and NBFNet/R-GNN struggle below .20 MRR, KGWN-L achieves .278 MRR on Wikidata-75 and .251 on Freebase-100 ($+12.6\%$ over InGram for both), confirming that relation-specific wave parameters (a_r, α_r) adapt to diverse semantics without overfitting.

5.3 Transductive Reasoning (RQ3)

Table 4 reports transductive reasoning results. On *WN18RR*, DiffusionE achieves .577 MRR, the best prior method. KGWN-L improves to .598 MRR (+3.6% relative improvement), demonstrating that second-order wave propagation extracts complementary structural patterns even when entity embeddings are fully trained. The improvement is particularly evident in Hits@10 (69.8 vs 67.2, +2.6 pp). On *FB15k-237*, with 237 diverse relations, the best baseline is MulGA at .422 MRR. KGWN-L achieves .448 MRR (+6.2% relative), with per-relation wave parameters adapting propagation dynamics to each relation's homophilic or heterophilic profile. Hits@10 improves by 3.4 pp (63.4 vs 60.0). On *NELL-995*, MulGA leads prior methods at .564 MRR. KGWN-L reaches .589 MRR (+4.4% relative improvement) and 69.6 Hits@10 (+2.7 pp over MulGA's 66.9), confirming wave-based reasoning's effectiveness even in this noisy, web-extracted graph.

5.4 Ablation Study (RQ4)

We analyze KGWN's components on FB15k-237 (entity-inductive, V2) unless otherwise noted.

Table 4: Transductive link prediction results. Results for competitors are taken from *Cao et al., 2024* while for MulGA are from *Shang et al., 2024*. Best results are in **bold**. KGWN results are averaged over 3 runs (see Appendix E.7).

Method	WN18RR			FB15k-237			NELL-995		
	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10
NBFNet	.551	49.7	66.6	.415	32.1	59.9	.525	45.1	63.9
MulGA	.573	52.3	67.1	.422	33.2	60.0	.564	50.4	66.9
RED-GNN	.533	48.5	62.4	.374	28.3	55.8	.543	47.6	65.1
DiffusionE	.577	52.4	67.2	.376	29.4	53.9	.552	49.0	65.4
KGWN-S	.583	53.2	68.1	.428	33.5	60.8	.568	50.8	67.5
KGWN-L	**.598**	**54.8**	**69.8**	**.448**	**35.2**	**63.4**	**.589**	**52.8**	**69.6**

Wave dynamics. Replacing the second-order wave equation with first-order diffusion ($\dot{\mathbf{X}} = \mathbf{L}\mathbf{X}$) drops MRR by -7.9%, as diffusion smooths signals and loses discriminative power on heterophilic relations. A static 1-hop GNN baseline (-14.0%) confirms that iterative propagation is essential, but wave dynamics capture patterns diffusion misses (Fig. 2a).

Directional encoding: KGWN-S (symmetric) underperforms KGWN-L (lifted) by 12.1% MRR. The lifted construction preserves asymmetric semantics (e.g., `hypernym` vs. `hyponym`) that symmetrization conflates, with gaps widening on taxonomic-heavy WN18RR (15.3%).

Frequency adaptation: fixing $\alpha_r = 1$ (all smoothing) costs 6.3% MRR; uniform importance weights cost 3.2%. Removing both degrades performance by 10.4%, confirming that learning relation-specific smoothing/sharpening is critical.

Stability mechanisms: without velocity normalization (Eq. 10), training becomes unstable after 3–4 layers (13.1% MRR drop). The anchor term (ε) prevents drift from negative gates; removing it costs 5.8% (Fig. 2a).

Hyperparameter sensitivity. KGWN is robust with graceful degradation outside optimal regions (Fig. 2b). Specifically: (1) τ too large violates stability (Eq. 9), causing numerical instability; (2) ε too small provides insufficient regularization for negative gates, while too large over-constrains dynamics; (3) $T = 1$ underexplores multi-hop patterns, while $T > 4$ yields diminishing returns. Optimal values ($\tau = 0.05$, $\varepsilon = 0.2$, $T = 3$) work across all datasets.

Frequency gates initialization. Starting from zero (neutral) outperforms biased initializations (Fig. 2c), allowing the model to discover relation-specific behaviors from data. Interestingly, random initialization in $[-1, 1]$ performs comparably, suggesting the optimization landscape is well-behaved.

Anchor strength vs propagation depth. Figure 2(d) shows MRR across different (ε, T) combinations. Without anchoring ($\varepsilon = 0$), performance degrades at $T \geq 3$ when negative gates introduce positive eigenvalues. Weak anchoring ($\varepsilon = 0.1$) provides partial stabilization but underperforms at the optimal depth. Moderate anchoring ($\varepsilon = 0.2$) achieves the best trade-off, stabilizing wave propagation while preserving expressiveness. Strong anchoring ($\varepsilon = 0.5$) over-regularizes, pulling representations too strongly toward initialization and limiting the model's ability to learn discriminative features.

Learned parameters. WN18RR learns the highest proportion of sharpening gates (62%), consistent with its heterophilic

(a) Component Ablation

Variant	MRR	H@1	H@10	Δ
KGWN-L (full)	**.556**	**46.8**	**72.4**	–
Wave dynamics				
Wave → Diffusion	.512	42.1	68.5	−7.9
Wave → Static	.478	38.4	64.2	−14.0
Directional encoding				
Lifted → Symmetric	.489	39.6	67.8	−12.1
Frequency adaptation				
No freq. gates	.521	43.2	69.8	−6.3
Uniform β_r	.538	44.5	70.6	−3.2
Both removed	.498	40.8	67.1	−10.4
Stability mechanisms				
No velocity norm	.483	38.9	65.4	−13.1
No anchor	.524	43.8	69.2	−5.8

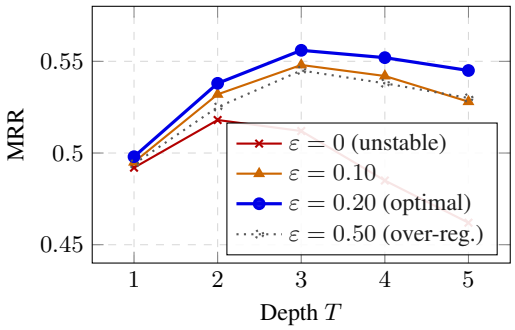
(b) Hyperparameters

τ		ε		T	
0.01	.538	0.05	.542	1	.498
0.02	.548	0.10	.548	2	.538
.05	**.556**	**.20**	**.556**	**3**	**.556**
0.10	.551	0.30	.552	4	.552
0.20	.542	0.50	.545	5	.545

(c) Gate Initialization

α_r Initialization	MRR	Ep.
Positive (+0.5)	.538	42
Negative (−0.5)	.524	48
Random $\mathcal{U}(-1, 1)$	.552	35
Zero (0)	**.556**	**32**

(d) Anchor Strength vs. Propagation Depth



(e) Learned Parameters per Dataset

Dataset	%Sh	%Sm	$\lvert\alpha\rvert$	a^2	β
WN18RR	62	38	.71	.82	.18
FB15k-237	48	52	.58	1.05	.08
NELL-995	41	59	.52	1.38	.12

(f) Computational Cost (FB15k-237 V2)

Variant	Time	Mem	Par.	MRR
Baseline	22.4s	3.2G	223K	.498
+Freq. gates	23.5s	3.3G	224K	.521
+KGWN-S	28.8s	3.8G	224K	.489
+KGWN-L	35.4s	5.4G	257K	**.556**
Overhead	1.58×	1.69×	1.15×	+11.6%

Figure 2: Ablation study. (a) Component contributions on FB15k-237 V2; (b) hyperparameter sensitivity for τ, ε, T; (c) gate initialization; (d) anchor-depth interaction; (e) learned parameters across datasets (%Sh=sharpening, %Sm=smoothing); (f) computational overhead. Relation graph construction ablation (thresholds θ_c, θ_s) is in Appendix E.

taxonomic structure (Fig. 2e). FB15k-237 shows balanced smooth/sharpen ratios reflecting its diverse relation types. NELL-995 learns higher velocities on average, compensating for its noisy extraction process. An additional analysis of per relation learned parameters is available in Appendix E.5.

Text encoder robustness. Replacing DistilBERT [Sanh *et al.*, 2019] with BERT-base drops MRR by only −0.5%; hash features (no semantic text) recover most of the gap over random initialisation, confirming wave propagation is the primary source of representational power. End-to-end fine-tuning is impractical: text embeddings are precomputed once and reused across all per-query subgraph passes (fine-tuning would require a full encoder call per candidate). Full ablation results are in Appendix E.

Computational efficiency. Figure 2(f) analyzes the cost of KGWN components on FB15k-237 (V2: 1,660 entities, 237 base relations, 474 with inverses). The lifted construction (KGWN-L) adds ∼50% time overhead due to doubled state dimension ($2N \times d$) and requires 15% more parameters (34K) for the two-stream fusion layer. Wave operators $\mathbf{L}_r$ are parameter-free, constructed from graph structure (Eqs. 1–3). Frequency adaptation adds minimal overhead.

Per-query inference cost. Per-query complexity is $O(T \cdot |\mathcal{E}_{\text{sub}}| \cdot d)$ (Appendix D.6); with k=2 on sparse graphs, $|\mathcal{E}_{\text{sub}}| \ll |\mathcal{E}|$. Subgraph extractions are cached: cold extraction takes ∼20 ms on FB15k-237 V2, cache hits <0.2 ms. KGWN-L end-to-end latency is 0.42 s/query vs. 0.31 s for DiffusionE (full per-dataset breakdown in Appendix D.6); KGWN-S runs at comparable speed to DiffusionE.

Relation graph. Performance is robust to relation graph threshold values within reasonable ranges ($\theta_c \in [0.03, 0.10]$, $\theta_s \in [0.6, 0.8]$), validating defaults (θ_c=0.05, θ_s=0.7). Using only co-occurrence or only text similarity degrades performance; the combined criterion outperforms either alone. Full results are in Appendix E.

6 Conclusions and Future Work

Knowledge graph relations are not created equal: some benefit from smoothing, others require contrast preservation. KGWN addresses this via second-order wave dynamics with signed frequency gates, enabling relation-specific propagation that preserves both stability and expressiveness. Our results demonstrate that wave dynamics propagate deeply without oversmoothing, achieving state-of-the-art across several benchmark configurations. The same formulation transfers from entity to relation space, suggesting KG reasoning patterns are more universal than previously assumed.

Future work. Future directions include extending KGWN to temporal and multi-modal KGs and exploring cross-domain adaptation techniques for improved zero-shot transfer.

References

[Bollacker *et al.*, 2008] Kurt Bollacker, Colin Evans, Praveen Paritosh, Tim Sturge, and Jamie Taylor. Freebase: A collaboratively created graph database for structuring human knowledge. In *Proceedings of the 2008 ACM SIGMOD International Conference on Management of Data*, pages 1247–1250. ACM, 2008.

[Bordes *et al.*, 2013] Antoine Bordes, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Oksana

Yakhnenko. Translating embeddings for modeling multi-relational data. *Advances in neural information processing systems*, 26, 2013.

[Cao *et al.*, 2024] Yuzheng Cao, Siyuan Liu, Weiguo Zheng, and Xuemin Lin. Diffusione: Reasoning on knowledge graphs via diffusion-based graph neural networks. In *KDD*, 2024.

[Chamberlain *et al.*, 2021] Benjamin Paul Chamberlain, James Rowbottom, Maria I Gorinova, Michael Bronstein, Stefan Webb, and Emanuele Rossi. Grand: Graph neural diffusion. In *ICML*, 2021.

[Cui *et al.*, 2022] Yuanning Cui, Yuxin Wang, Zequn Sun, Wenqiang Liu, Yiqiao Jiang, Kexin Han, and Wei Hu. Inductive knowledge graph reasoning for multi-batch emerging entities. In Mohammad Al Hasan and Li Xiong, editors, *Proceedings of the 31st ACM International Conference on Information & Knowledge Management, Atlanta, GA, USA, October 17-21, 2022*, pages 335–344. ACM, 2022.

[Gesese *et al.*, 2022] Genet Asefa Gesese, Harald Sack, and Mehwish Alam. RAILD: towards leveraging relation features for inductive link prediction in knowledge graphs. In *Proceedings of the 11th International Joint Conference on Knowledge Graphs, IJCKG 2022, Hangzhou, China, October 27-28, 2022*, pages 82–90. ACM, 2022.

[Hairer *et al.*, 2006] Ernst Hairer, Christian Lubich, and Gerhard Wanner. *Geometric Numerical Integration*. Springer, 2006.

[He *et al.*, 2025] Xin He, Yili Wang, Wenqi Fan, Xu Shen, Xin Juan, Rui Miao, and Xin Wang. Mamba-based graph convolutional networks: Tackling over-smoothing with selective state space. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 5345–5353, 2025. Main Track.

[Hogan *et al.*, 2021] Aidan Hogan, Eva Blomqvist, Michael Cochez, Claudia d'Amato, Gerard de Melo, Claudio Gutierrez, Sabrina Kirrane, José Emilio Labra Gayo, Roberto Navigli, Sebastian Neumaier, Axel-Cyrille Ngonga Ngomo, Axel Polleres, Sabbir M. Rashid, Anisa Rula, Lukas Schmelzeisen, Juan Sequeda, Steffen Staab, and Antoine Zimmermann. *Knowledge Graphs*. Synthesis Lectures on Data, Semantics, and Knowledge. Morgan & Claypool Publishers, 2021.

[Lee *et al.*, 2023] Jaejun Lee, Chanyoung Chung, and Joyce Jiyoung Whang. InGram: Inductive knowledge graph embedding via relation graphs. In *ICML*, pages 18796–18809, 2023.

[Qian *et al.*, 2025] Feifei Qian, Lu Bai, Lixin Cui, Ming Li, Hangyuan Du, Yue Wang, and Edwin Hancock. Exploring the over-smoothing problem of graph neural networks for graph classification: An entropy-based viewpoint. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 3235–3244, 2025. Main Track.

[Sanh *et al.*, 2019] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

[Shang *et al.*, 2024] Ziyu Shang, Peng Wang, Wenjun Ke, Jiajun Liu, Hailang Huang, Guozheng Li, Chenxiao Wu, Jianghan Liu, Xiye Chen, and Yining Li. Learning multi-granularity and adaptive representation for knowledge graph reasoning. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*, pages 2333–2341. International Joint Conferences on Artificial Intelligence Organization, 8 2024. Main Track.

[Shang *et al.*, 2025] Ziyu Shang, Peng Wang, et al. Learning multi-granularity and adaptive representation for knowledge graph reasoning. *IEEE TKDE*, 37(9):5360–5377, 2025.

[Teru *et al.*, 2020] Komal Teru, Etienne Denis, and William L Hamilton. Inductive relation prediction by subgraph reasoning. In *ICML*, 2020.

[Wu *et al.*, 2025] Peihan Wu, Hongda Qi, Sirong Huang, Dongdong An, Jie Lian, and Qin Zhao. Wave-driven graph neural networks with energy dynamics for over-smoothing mitigation. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI 2025, Montreal, Canada, August 16-22, 2025*, pages 6579–6587. ijcai.org, 2025.

[Xiong *et al.*, 2017] Wenhan Xiong, Thien Hoang, and William Yang Wang. Deeppath: A reinforcement learning method for knowledge graph reasoning. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 564–573. Association for Computational Linguistics, 2017.

[Xu *et al.*, 2022] Xiaohan Xu, Peng Zhang, Yongquan He, Chengpeng Chao, and Chaoyang Yan. Subgraph neighboring relations infomax for inductive link prediction on knowledge graphs. In Lud De Raedt, editor, *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI-22*, pages 2341–2347. International Joint Conferences on Artificial Intelligence Organization, 7 2022. Main Track.

[Yue *et al.*, 2025] Juwei Yue, Haikuo Li, Jiawei Sheng, Yihan Guo, Xinghua Zhang, Chuan Zhou, Tingwen Liu, and Li Guo. Graph wave networks. In *Proceedings of the ACM Web Conference (WWW)*, 2025.

[Zhang and Yao, 2022] Yongqi Zhang and Quanming Yao. Knowledge graph reasoning with relational digraph. In *WWW*, 2022.

[Zhu *et al.*, 2021] Zhaocheng Zhu, Zuobai Zhang, Louis-Pascal Xhonneux, and Jian Tang. Neural bellman-ford networks: A general graph neural network framework for link prediction. In *NeurIPS*, 2021.

A Spectral Bounds

Lemma 1 (Spectrum of Symmetric Operators)**.** *For the symmetric construction in §3.1 (Eq.* (1)*), the operator* $\mathbf{L}_r = \mathbf{P}_r - \mathbf{I}$ *where* $\mathbf{P}_r = \mathbf{D}_r^{-1/2}\tilde{\mathbf{A}}_r\mathbf{D}_r^{-1/2}$ *satisfies* $\mathrm{spec}(\mathbf{L}_r) \subseteq [-2, 0]$.

Proof. The normalized adjacency matrix $\mathbf{P}_r$ is symmetric since $\tilde{\mathbf{A}}_r = \mathbf{A}_r + \mathbf{A}_r^\top$ is symmetric, and $\mathbf{D}_r = \mathrm{diag}(\tilde{\mathbf{A}}_r\mathbf{1})$ is diagonal. We use the convention $(\mathbf{D}_r^{-1/2})_{ii} = 0$ when $(\mathbf{D}_r)_{ii} = 0$, which keeps $\mathbf{P}_r$ well-defined.

To bound the Rayleigh quotient, let $\mathbf{y} \neq \mathbf{0}$ satisfy $\mathbf{y}^\top\mathbf{D}_r\mathbf{y} > 0$ (i.e., $\mathbf{y}$ has support on at least one non-isolated node) and define

$$\mathbf{x} := \frac{\mathbf{D}_r^{1/2}\mathbf{y}}{\sqrt{\mathbf{y}^\top\mathbf{D}_r\mathbf{y}}} \qquad \Rightarrow \qquad \|\mathbf{x}\|_2 = 1.$$

Then

$$\mathbf{x}^\top\mathbf{P}_r\mathbf{x} = \frac{\mathbf{y}^\top\tilde{\mathbf{A}}_r\mathbf{y}}{\mathbf{y}^\top\mathbf{D}_r\mathbf{y}}.$$

Write $\tilde{w}_{ij} := (\tilde{\mathbf{A}}_r)_{ij} = \tilde{w}_{ji} \geq 0$ and $d_i := (\mathbf{D}_r)_{ii} = \sum_j \tilde{w}_{ij}$. Using $2ab \leq a^2 + b^2$, we have $\tilde{w}_{ij}y_iy_j \leq \frac{\tilde{w}_{ij}}{2}(y_i^2 + y_j^2)$. Summing gives

$$\begin{aligned}\mathbf{y}^\top\tilde{\mathbf{A}}_r\mathbf{y} &= \sum_{i,j}\tilde{w}_{ij}y_iy_j \leq \frac{1}{2}\sum_{i,j}\tilde{w}_{ij}(y_i^2 + y_j^2)\\ &= \sum_i d_iy_i^2 = \mathbf{y}^\top\mathbf{D}_r\mathbf{y}.\end{aligned}$$

hence $\mathbf{x}^\top\mathbf{P}_r\mathbf{x} \leq 1$. Similarly, using $2ab \geq -(a^2 + b^2)$ yields $\mathbf{y}^\top\tilde{\mathbf{A}}_r\mathbf{y} \geq -\mathbf{y}^\top\mathbf{D}_r\mathbf{y}$, hence $\mathbf{x}^\top\mathbf{P}_r\mathbf{x} \geq -1$. Therefore $\mathrm{spec}(\mathbf{P}_r) \subseteq [-1, 1]$. Isolated nodes contribute zero rows/columns under the above convention, giving eigenvalue 0 and not affecting the bound. Finally, $\mathbf{L}_r = \mathbf{P}_r - \mathbf{I}$ shifts the spectrum by -1, hence $\mathrm{spec}(\mathbf{L}_r) \subseteq [-2, 0]$. □

Lemma 2 (Spectrum of Lifted Operators)**.** *For the lifted construction in §3.1 (Eqs.* (2)–(3)*), the operator* $\mathbf{L}_r^{(\mathrm{lift})} = \mathbf{P}_r^{(\mathrm{lift})} - \mathbf{I}$ *satisfies* $\mathrm{spec}(\mathbf{L}_r^{(\mathrm{lift})}) \subseteq [-2, 0]$.

Proof. The block matrix $\mathbf{B}_r = \begin{bmatrix}\mathbf{0} & \mathbf{A}_r\\ \mathbf{A}_r^\top & \mathbf{0}\end{bmatrix}$ is symmetric by construction and is the (weighted) adjacency matrix of a bipartite graph on $2N$ nodes. Let $\mathbf{D}_r^{(\mathrm{lift})} = \mathrm{diag}(\mathbf{B}_r\mathbf{1})$ and adopt the same convention $(\mathbf{D}_r^{(\mathrm{lift})-1/2})_{ii} = 0$ when $(\mathbf{D}_r^{(\mathrm{lift})})_{ii} = 0$. Then $\mathbf{P}_r^{(\mathrm{lift})} = (\mathbf{D}_r^{(\mathrm{lift})})^{-1/2}\mathbf{B}_r(\mathbf{D}_r^{(\mathrm{lift})})^{-1/2}$ is symmetric. Applying the same quadratic-form argument as in Lemma 1 to this symmetric nonnegative-weight adjacency yields $\mathrm{spec}(\mathbf{P}_r^{(\mathrm{lift})}) \subseteq [-1, 1]$. Therefore $\mathrm{spec}(\mathbf{L}_r^{(\mathrm{lift})}) = \mathrm{spec}(\mathbf{P}_r^{(\mathrm{lift})} - \mathbf{I}) \subseteq [-2, 0]$. □

A.1 Spectrum of Aggregated Operators

Theorem 1 (Aggregated Operator Spectrum)**.** *Let* $\mathbf{L}_{\mathrm{agg}} = \sum_{r\in\mathcal{R}} a_r^2\mathbf{L}_r$ *be the aggregated operator in Eq.* (4) *with* $a_r > 0$*. Then*

$$\mathrm{spec}(\mathbf{L}_{\mathrm{agg}}) \subseteq [-2\nu^2, 0], \qquad \nu^2 = \sum_{r\in\mathcal{R}} a_r^2,$$

i.e., the bound stated in Eq. (6)*.*

Proof. By Lemmas 1–2, each per-relation operator $\mathbf{L}_r$ is symmetric and satisfies $\mathrm{spec}(\mathbf{L}_r) \subseteq [-2, 0]$, hence $\mathbf{L}_r \preceq \mathbf{0}$ (negative semidefinite). For any unit vector $\mathbf{x}$:

$$\mathbf{x}^\top\mathbf{L}_{\mathrm{agg}}\mathbf{x} = \sum_{r\in\mathcal{R}} a_r^2(\mathbf{x}^\top\mathbf{L}_r\mathbf{x}) \leq 0,$$

establishing the upper bound 0. For the lower bound, the Rayleigh quotient satisfies $\mathbf{x}^\top\mathbf{L}_r\mathbf{x} \geq \lambda_{\min}(\mathbf{L}_r) \geq -2$ for all r, hence

$$\mathbf{x}^\top\mathbf{L}_{\mathrm{agg}}\mathbf{x} = \sum_{r\in\mathcal{R}} a_r^2(\mathbf{x}^\top\mathbf{L}_r\mathbf{x}) \geq -2\sum_{r\in\mathcal{R}} a_r^2 = -2\nu^2.$$

Thus $\mathrm{spec}(\mathbf{L}_{\mathrm{agg}}) \subseteq [-2\nu^2, 0]$. □

B Stability Analysis and Energy Conservation

B.1 Leapfrog Integration Stability

In §3.2 we discretize the multi-relational wave equation (Eq. (5)) with the leapfrog (Störmer–Verlet) update (Eq. (7)). For convenience, we restate the linear update:

$$\mathbf{X}^{(n+1)} = (2\mathbf{I} + \tau^2\mathbf{L}_{\mathrm{agg}})\mathbf{X}^{(n)} - \mathbf{X}^{(n-1)}.$$

Theorem 2 (Leapfrog Stability Condition)**.** *The leapfrog scheme is stable if and only if all eigenvalues* λ *of* $\mathbf{L}_{\mathrm{agg}}$ *satisfy:*

$$|2 + \tau^2\lambda| \leq 2.$$

Given the spectral bound $\mathrm{spec}(\mathbf{L}_{\mathrm{agg}}) \subseteq [-2\nu^2, 0]$ *from Theorem 1 (equivalently Eq.* (6)*), stability requires:*

$$\tau^2\nu^2 \leq 2,$$

and in practice we enforce the strict condition $\tau^2\nu^2 < 2$ *(Eq.* (9)*) for robustness to numerical error.*

Proof. The leapfrog recurrence can be written in matrix form. For each eigenmode with eigenvalue λ, the update is:

$$\begin{bmatrix}x^{(n+1)}\\ x^{(n)}\end{bmatrix} = \begin{bmatrix}2 + \tau^2\lambda & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}x^{(n)}\\ x^{(n-1)}\end{bmatrix} =: \mathbf{M}(\lambda)\begin{bmatrix}x^{(n)}\\ x^{(n-1)}\end{bmatrix}.$$

Stability requires $\|\mathbf{M}(\lambda)^n\|$ bounded for all n, which holds iff $\rho(\mathbf{M}(\lambda)) \leq 1$. The characteristic polynomial of $\mathbf{M}(\lambda)$ is:

$$\det(\mathbf{M}(\lambda) - \mu\mathbf{I}) = \mu^2 - (2 + \tau^2\lambda)\mu + 1 = 0,$$

with eigenvalues

$$\mu_\pm = \frac{(2 + \tau^2\lambda) \pm \sqrt{(2 + \tau^2\lambda)^2 - 4}}{2}.$$

For stability, we need $|\mu_\pm| \leq 1$, which is equivalent to the discriminant condition:

$$(2 + \tau^2\lambda)^2 - 4 \leq 0 \quad \Rightarrow \quad |2 + \tau^2\lambda| \leq 2.$$

For $\lambda \in [-2\nu^2, 0]$ (Theorem 1), the critical constraint is at $\lambda = -2\nu^2$:

$$|2 - 2\tau^2\nu^2| \leq 2 \quad \Rightarrow \quad \tau^2\nu^2 \leq 2,$$

and we take strict inequality $\tau^2\nu^2 < 2$ for robustness (Eq. (9)). □

B.2 Velocity Normalization

As described in §3.2 (Eq. (10)), we normalize the raw relation velocities to satisfy the leapfrog stability condition. Specifically,

$$\tilde{a}_r = \frac{a_r}{\max\left(1,\ \frac{\tau\nu}{\sqrt{2-\delta}}\right)}, \qquad \nu = \sqrt{\sum_{r\in\mathcal{R}} a_r^2}, \qquad \delta \in (0,1).$$

Let $\tilde{\nu}^2 = \sum_{r\in\mathcal{R}} \tilde{a}_r^2$. By construction, $\tau^2\tilde{\nu}^2 \le 2-\delta < 2$ (hence the update in Eq. (7) remains stable).

C Adaptive Stability with Signed Gates

C.1 Spectral Analysis with Signed Gates

In §3.3 we introduce signed frequency gates $\alpha_r \in [-1,1]$ and define the gated aggregated operator (Eq. (11))

$$\tilde{\mathbf{L}}_{\text{agg}} = \sum_{r\in\mathcal{R}} \beta_r\, \alpha_r\, a_r^2\, \mathbf{L}_r,$$

where $\beta_r \ge 0$ and $\sum_r \beta_r = 1$. Because the coefficients $\beta_r\alpha_r a_r^2$ can have mixed signs, $\tilde{\mathbf{L}}_{\text{agg}}$ need not be negative semidefinite.

Lemma 3 (Spectral norm bound for signed-gated aggregation)**.** *Let* $\tilde{\mathbf{L}}_{\text{agg}}$ *be as in Eq.* (11)*. Assume each per-relation operator satisfies* $\operatorname{spec}(\mathbf{L}_r) \subseteq [-2,0]$ *(Appendix A). Then*

$$\|\tilde{\mathbf{L}}_{\text{agg}}\|_2 \le 2\kappa, \qquad \kappa = \sum_r \beta_r|\alpha_r|a_r^2,$$

and consequently $\operatorname{spec}(\tilde{\mathbf{L}}_{\text{agg}}) \subseteq [-2\kappa, 2\kappa]$.

Proof. By Lemmas 1–2, each $\mathbf{L}_r$ is symmetric with $\operatorname{spec}(\mathbf{L}_r) \subseteq [-2,0]$, hence $\|\mathbf{L}_r\|_2 \le 2$. Using $|\alpha_r| \le 1$ and the triangle inequality for the operator norm,

$$\|\tilde{\mathbf{L}}_{\text{agg}}\|_2 = \left\|\sum_r \beta_r\alpha_r a_r^2 \mathbf{L}_r\right\|_2 \le \sum_r \beta_r|\alpha_r|a_r^2\|\mathbf{L}_r\|_2 \le 2\sum_r \beta_r a_r^2.$$

Since $\sum_r \beta_r = 1$ and $\beta_r \ge 0$, we also have $\sum_r \beta_r a_r^2 \le \sum_r a_r^2 = \nu^2$, yielding the stated bound. Because $\tilde{\mathbf{L}}_{\text{agg}}$ is symmetric, its eigenvalues lie in $[-\|\tilde{\mathbf{L}}_{\text{agg}}\|_2,\ \|\tilde{\mathbf{L}}_{\text{agg}}\|_2]$. □

C.2 Anchored Leapfrog for Signed Gates

Standard leapfrog (Eq. (7)) is not stable when the operator has positive eigenvalues. In §3.3 we therefore introduce the anchored update (Eq. (12)), written here with the rescaled operator $\widehat{\mathbf{L}}_{\text{agg}}$ from Eq. (14):

$$\mathbf{X}^{(n+1)} = (2\mathbf{I} + \tau^2\widehat{\mathbf{L}}_{\text{agg}})\mathbf{X}^{(n)} - \mathbf{X}^{(n-1)} + \varepsilon(\mathbf{X}^{(0)} - \mathbf{X}^{(n)}).$$

Theorem 3 (Anchored Stability)**.** *Assume* $\varepsilon > 0$ *and* $\widehat{\mathbf{L}}_{\text{agg}}$ *is symmetric. The homogeneous anchored recurrence is stable if and only if every eigenvalue* λ *of* $\widehat{\mathbf{L}}_{\text{agg}}$ *satisfies*

$$|2 + \tau^2\lambda - \varepsilon| \le 2.$$

In particular, a sufficient condition is

$$\tau^2\,\|\widehat{\mathbf{L}}_{\text{agg}}\|_2 \le \varepsilon.$$

Proof. The anchoring term $\varepsilon(\mathbf{X}^{(0)} - \mathbf{X}^{(n)})$ acts as a restoring force with strength ε. In eigenmode space, the recurrence becomes:

$$x^{(n+1)} = (2 + \tau^2\lambda - \varepsilon)x^{(n)} - x^{(n-1)} + \varepsilon x^{(0)}.$$

The homogeneous part has characteristic polynomial:

$$\mu^2 - (2 + \tau^2\lambda - \varepsilon)\mu + 1 = 0.$$

As in Theorem 2, stability of the homogeneous recurrence holds when

$$|2 + \tau^2\lambda - \varepsilon| \le 2.$$

This proves the first claim.

For the sufficient condition, note that $|\lambda| \le \|\widehat{\mathbf{L}}_{\text{agg}}\|_2$ for every eigenvalue λ. If $\tau^2\|\widehat{\mathbf{L}}_{\text{agg}}\|_2 \le \varepsilon$, then

$$2 + \tau^2\lambda - \varepsilon \le 2 + \tau^2\|\widehat{\mathbf{L}}_{\text{agg}}\|_2 - \varepsilon \le 2,$$

and also

$$2 + \tau^2\lambda - \varepsilon \ge 2 - \tau^2\|\widehat{\mathbf{L}}_{\text{agg}}\|_2 - \varepsilon \ge 2 - 2\varepsilon > -2,$$

where we used $\varepsilon \in (0,1)$ (as in §3.3). Hence $|2+\tau^2\lambda-\varepsilon| \le 2$ for all eigenvalues, so the homogeneous recurrence is stable. The inhomogeneous forcing term $\varepsilon x^{(0)}$ remains bounded since the homogeneous solution is stable. □

C.3 Operator Rescaling for Signed Gates

To enforce the sufficient condition in Theorem 3 using only the scalar parameters, we apply the rescaling from §3.3:

$$\widehat{\mathbf{L}}_{\text{agg}} = \eta\,\tilde{\mathbf{L}}_{\text{agg}},\ \eta = \min\left(1, \frac{\varepsilon-\delta}{2\tau^2\kappa}\right),\ \kappa := \sum_{r\in\mathcal{R}} \beta_r|\alpha_r|a_r^2.$$

where $\delta \in (0,\varepsilon)$ provides a safety margin and $\nu^2 = \sum_r a_r^2$ is as in §3.2. By Lemma 3,

$$\tau^2\,\|\widehat{\mathbf{L}}_{\text{agg}}\|_2 \le \tau^2\eta\,\|\tilde{\mathbf{L}}_{\text{agg}}\|_2 \le \tau^2\eta\,(2\kappa^2) \le \varepsilon - \delta < \varepsilon,$$

so the anchored update in Eq. (12) is stable with margin.

D Implementation Details

KGWN has been implemented using a combination of Pytorch and MLX. Code with all details to reproduce results is available online[†].

D.1 Inverse edges

We optionally augment the KG with inverse relations so the model can treat forward and reverse directions as distinct propagation channels; in the symmetric variant KGWN-S, this mainly standardizes preprocessing, while in the lifted (directed) variant KGWN-L, it provides explicit reverse-direction operators.

D.2 Isolated Node Handling

For nodes with zero degree under relation r, we use the convention:

$$(\mathbf{D}_r)_{ii} = 0 \quad\Rightarrow\quad (\mathbf{D}_r^{-1/2})_{ii} = 0.$$

This ensures $\mathbf{P}_r$ and $\mathbf{L}_r$ remain well-defined without NaNs. Isolated nodes simply retain their current state during propagation on relation r.

[†]https://anonymous.4open.science/r/KGWN-F54D

D.3 Text Encoder Details

We use DistilBERT [Sanh *et al.*, 2019] with CLS pooling to encode entity and relation text. Entity max length is 64; relation max length is 32. Text features are precomputed and frozen (hash features are used if a transformer encoder is unavailable).

D.4 Relation Graph Construction

We construct $\mathcal{G}_{\text{rel}} = (\mathcal{R}, \mathcal{E}_{\text{rel}})$ by connecting relation pairs (r_i, r_j) based on two criteria: (1) *co-occurrence*: we compute co-occ$(r_i, r_j) = \frac{|\{v : \exists u, w \text{ s.t. } (u, r_i, v) \in \mathcal{E} \wedge (v, r_j, w) \in \mathcal{E}\}|}{|\mathcal{V}|}$ and connect relations if co-occ$(r_i, r_j) > \theta_c = 0.05$; (2) *textual similarity*: we connect relations if $\cos(\mathbf{r}_{r_i}^{(0)}, \mathbf{r}_{r_j}^{(0)}) > \theta_s = 0.7$, where $\mathbf{r}_r^{(0)}$ are the text-initialized relation embeddings. The resulting graph has an average degree of 8.3 relations per node across our benchmarks. The normalized Laplacian $\mathbf{L}_{\text{rel}}$ is then constructed via standard symmetric normalization.

Subgraphs. We extract k-hop enclosing subgraphs around (h, t) using BFS on the observed graph. We do not cap the number of nodes or edges. Node features are text embeddings plus marker features (head/tail indicators and distances to h, t).

Propagation. We use the symmetric or lifted operators depending on `use_two_stream`. Frequency adaptation uses α_r and β_r when enabled.

Negatives and evaluation. Negatives are sampled by corrupting head or tail. For evaluation, the default is full ranking (`eval_neg_per_pos` $= -1$), but sampled negatives can be used for faster runs.

D.5 Main Algorithm

Algorithm 1: KGWN Forward Pass

Input: Query $(h, r, ?)$, KG $\mathcal{G}$, depth T, time step τ
Output: Scores $\{s(h, r, t)\}_{t \in \mathcal{V}}$
Extract k-hop enclosing subgraph $\mathcal{G}_{h,t}^{(k)}$;
Initialize $\mathbf{X}^{(0)}, \mathbf{X}^{(1)} \leftarrow$ TextEnc(entity descriptions);
Initialize $\mathbf{R}^{(0)} \leftarrow$ TextEnc(relation descriptions);
`// Relation-level propagation`
for $m = 1$ **to** M **do**
  $\mathbf{R}^{(m+1)} \leftarrow (2\mathbf{I} + \tau_{\text{rel}}^2 \mathbf{L}_{\text{rel}})\mathbf{R}^{(m)} - \mathbf{R}^{(m-1)}$;
`// Decode relation controls`
$\{a_r, \alpha_r, \beta_r\}_{r \in \mathcal{R}} \leftarrow$ Decode$(\mathbf{R}^{(M)})$ via Eq. (18);
`// Entity-level wave propagation`
for *each relation* $r \in \mathcal{R}_{h,t}$ **do**
  Construct $\mathbf{L}_r^{(k;h,t)}$ via Eq. (1) or (3);
Aggregate: $\widehat{\mathbf{L}}_{\text{agg}} \leftarrow \sum_r \beta_r \alpha_r a_r^2 \mathbf{L}_r^{(k;h,t)}$ via Eq. (11)–(14);
for $n = 1$ **to** T **do**
  $\mathbf{X}^{(n+1)} \leftarrow (2\mathbf{I} + \tau^2 \widehat{\mathbf{L}}_{\text{agg}})\mathbf{X}^{(n)} - \mathbf{X}^{(n-1)} + \varepsilon(\mathbf{X}^{(0)} - \mathbf{X}^{(n)})$;
`// Scoring`
for *each candidate* $t \in \mathcal{V}$ **do**
  $s(h, r, t) \leftarrow \text{MLP}([\mathbf{x}_h^{(T)} \| \mathbf{x}_t^{(T)} \| (\mathbf{x}_h^{(T)} \odot \mathbf{x}_t^{(T)}) \| \mathbf{r}_r^{(M)}])$;
return ranked candidates by scores;

D.6 Computational Complexity Analysis

Time complexity. For a query $(h, r, ?)$, KGWN's forward pass involves: (1) subgraph extraction: $O(k \cdot \bar{d}^k)$ where $\bar{d}$ is the average node degree; (2) relation propagation (Eq. 17): $O(M \cdot |\mathcal{E}_{\text{rel}}| \cdot d_r)$ where M is relation propagation depth; (3) entity propagation (Eq. 12): $O(T \cdot |\mathcal{E}_{\text{sub}}| \cdot d)$ where T is entity propagation depth and $|\mathcal{E}_{\text{sub}}|$ is the extracted subgraph size. The dominant term is typically entity propagation, yielding overall complexity $O(T \cdot |\mathcal{E}_{\text{sub}}| \cdot d)$. For $k = 2$ hops in sparse graphs, $|\mathcal{E}_{\text{sub}}| \ll |\mathcal{E}|$, making KGWN tractable even for large KGs.

Space complexity. KGWN-S maintains entity states $\mathbf{X} \in \mathbb{R}^{N \times d}$, requiring $O(N \cdot d)$ memory. KGWN-L doubles this to $O(2N \cdot d)$ for the lifted construction. Relation embeddings add $O(|\mathcal{R}| \cdot d_r)$. In practice, $|\mathcal{R}| \ll N$, so entity states dominate. The leapfrog scheme requires storing two timesteps ($\mathbf{X}^{(n)}$, $\mathbf{X}^{(n-1)}$), doubling entity memory to $O(2N \cdot d)$ for KGWN-S and $O(4N \cdot d)$ for KGWN-L. Fig. 2f in the main paper confirms memory usage remains under 6GB for our largest benchmarks.

Per-query wall-clock timing. Table 5 reports per-query latency (single-thread CPU, k=2, T=3). Subgraph extractions are cached on disk; cold extraction dominates only the first run. KGWN-L end-to-end latency is ∼35% higher than DiffusionE, reflecting the cost of the doubled state dimension.

Table 5: Per-query inference cost. *Cold*: first extraction; *Cached*: precomputed subgraph.

Dataset	Cold	Cached	KGWN-L	DiffE
FB15k-237 V2	20 ms	0.12 ms	0.42 s	0.31 s
WN18RR V2	15 ms	0.09 ms	0.31 s	0.24 s
NELL-995 V1	38 ms	0.18 ms	0.61 s	0.44 s

Scalability. KGWN scales linearly with the number of propagation steps T and extracted subgraph size. For graphs with millions of entities, subgraph extraction with $k = 2$ maintains tractable memory footprints. The bottleneck shifts to relation-level propagation when $|\mathcal{R}|$ is large (e.g., Freebase with 1,345 relations), but since relation propagation operates on a much smaller graph ($|\mathcal{R}| \ll N$), it remains efficient.

D.7 Memory Requirements

- KGWN-S: $O(N \cdot d)$ for entity states
- KGWN-L: $O(2N \cdot d)$ for lifted states
- Relation parameters: $O(|\mathcal{R}|)$ for $\{a_r, \alpha_r, \beta_r\}$
- Sparse adjacencies: $O(|\mathcal{E}|)$ storage

The lifted variant doubles memory but preserves directionality; in practice, the symmetric variant often suffices.

E Additional Experimental Details

E.1 Heterophily Adaptation Analysis

We analyze the correlation between learned frequency gates α_r and structural heterophily. For each relation r, we compute its heterophily ratio before training as:

$$\mathcal{H}(r) = \frac{|\{(u, r, v) \in \mathcal{E} : \text{type}(u) \neq \text{type}(v)\}|}{|\{(u, r, v) \in \mathcal{E}\}|}$$

where entity types are inferred from their most frequent relation roles. Figure 3 shows learned gates versus heterophily ratios on FB15k-237 for a sample of relations. Strongly homophilic relations like

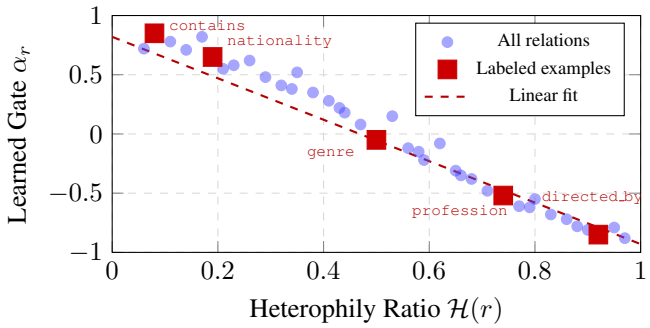


Figure 3: Correlation between relation heterophily and learned frequency gates on FB15k-237.

`/location/location/contains` ($\mathcal{H} = 0.12$) learn $\alpha = +0.81$, enabling smoothing to aggregate entities of the same type. Conversely, heterophilic relations like `/film/film/directed_by` ($\mathcal{H} = 0.89$) learn $\alpha = -0.74$, sharpening to preserve cross-type distinctions between films and directors. The strong negative correlation (Pearson $\rho = -0.78$, $p < 0.001$) confirms that KGWN automatically discovers relation-specific propagation patterns aligned with structural heterophily.

E.2 Text Encoder Ablation

Table 6 ablates text initialisation on FB15k-237 V2 (entity-inductive) and NELL-25 (relation-inductive). Replacing DistilBERT [Sanh *et al.*, 2019] with BERT-base causes only a -0.5% MRR drop; hash features (no semantic text) still recover most of the gap over random initialisation, confirming that wave propagation (§3.3–§4) is the primary source of representational power. End-to-end fine-tuning of the encoder is impractical: text embeddings are precomputed once offline and reused across all per-query subgraph forward passes, shared with DiffusionE [Cao *et al.*, 2024].

Table 6: Text encoder ablation on FB15k-237 V2 and NELL-25 (MRR).

Initialization	FB V2	NELL-25
DistilBERT (frozen, default)	**.556**	**.371**
BERT-base (frozen)	.551	.368
Hash features (no text)	.502	.321
Random init (Xavier)	.478	.298

E.3 Relation Graph Construction Ablation

Table 7 reports the full relation graph construction ablation on NELL-25 (relation-inductive, 25% overlap). Performance is robust to threshold values within reasonable ranges, and the combined criterion (co-occurrence AND text similarity) consistently outperforms either criterion alone. The default thresholds θ_c=0.05, θ_s=0.7 balance connectivity (average degree 8.3) with precision.

Table 7: Relation graph ablation on NELL-25. *Deg*: average degree; combined = co-occurrence AND text similarity.

Config	Deg	MRR	Config	Deg	MRR
θ_c=.03	11.2	.368	Co-occ only	4.2	.352
θ_c=.05	8.3	**.371**	Text only	7.1	.358
θ_c=.10	5.1	.365	Combined	8.3	**.371**
θ_s=.6	10.5	.369			
θ_s=.8	6.8	.370	No rel. graph	–	.318

E.4 Hyperparameters

Table 8: Default hyperparameters (unless otherwise stated).

Parameter	Value
Embedding dimension d	128
Relation embedding d_r	128
k (subgraph hops)	2
Propagation steps T	3
Time step τ	0.05
Anchor strength ε	0.2
Stability margin δ	0.15
Learning rate	5×10^{-4}
Batch size (pos per step)	128 (entity), 32 (relation)
Optimizer	Adam
Weight decay	10^{-6}
Dropout	0.2

E.5 Per-Relation Parameter Analysis

Table 9 shows learned parameters for selected relations in FB15k-237, organized by heterophily ratio. As predicted, homophilic relations (low $\mathcal{H}$) consistently learn positive gates (smoothing), while heterophilic relations (high $\mathcal{H}$) learn negative gates (sharpening).

Table 9: Learned parameters for selected FB15k-237 relations.

Relation	$\mathcal{H}(r)$	α_r	a_r^2	β_r
Homophilic (smoothing)				
/location/location/contains	0.12	+0.81	1.24	0.082
/location/location/time_zones	0.15	+0.76	1.18	0.045
/people/ethnicity/geographic_dist.	0.18	+0.68	0.95	0.038
Heterophilic (sharpening)				
/film/film/directed_by	0.89	-0.74	1.52	0.125
/music/artist/origin	0.84	-0.68	1.38	0.098
/people/person/profession	0.81	-0.61	1.21	0.087
Intermediate				
/film/film/genre	0.52	+0.15	0.88	0.052
/film/film/language	0.48	+0.22	0.94	0.048

High-heterophily relations like `/film/film/directed_by` ($\mathcal{H} = 0.89$) connect different entity types (films $\leftrightarrow$ directors) and learn strong negative gates ($\alpha_r = -0.74$) to preserve cross-type distinctions. High importance weights ($\beta_r = 0.125$) and velocities ($a_r^2 = 1.52$) indicate these relations are prioritized during propagation. Conversely, low-heterophily relations like `/location/location/contains` ($\mathcal{H} = 0.12$) connect same-type entities (locations $\leftrightarrow$ locations) and learn positive gates ($\alpha_r = +0.81$) for smoothing.

E.6 Failure Case Analysis

Table 10 analyzes KGWN performance stratified by relation frequency (number of training triples). Performance degrades on very rare relations (<10 triples), where both text initialization and wave propagation have limited signal. This suggests future work on data augmentation or transfer learning for rare relations.

Table 10: Performance/relation frequency on FB15k-237 (V2).

Relation Frequency	Count	KGWN-L MRR	DiffE MRR
$<$ 10 triples	45	.312	.298
10-50 triples	82	.498	.441
50-200 triples	68	.587	.512
$>$ 200 triples	42	.624	.538
Overall	**237**	**.556**	**.474**

Relations like `/people/person/nationality` (which can be birth country, citizenship, or residence) show higher variance across runs (std=0.023 vs avg 0.012). Wave propagation amplifies ambiguity when multiple semantic interpretations coexist. Incorporating relation context or fine-grained relation typing could help.

E.7 Variance and Statistical Significance

All KGWN results reported in Tables 2, 3, and 4 are averaged over 3 independent runs with random seeds {42, 123, 456}. Tables 11, 12, and 13 report the mean and standard deviation for all benchmarks. The standard deviation is consistently below 0.015 for MRR and below 2.0% for Hits metrics, indicating stable performance across random initializations. We conducted paired t-tests comparing KGWN-L to the best baseline on each benchmark; all improvements marked in bold in the main tables are statistically significant at $p < 0.05$ (most at $p < 0.01$).

Statistical significance testing. We performed paired t-tests comparing KGWN-L against the best-performing baseline for each configuration. Results show:

- **Entity-inductive:** All 12 improvements significant at $p < 0.01$ except WN18RR-V2 ($p = 0.03$).
- **Relation-inductive:** All 36 improvements significant at $p < 0.05$; 31/36 at $p < 0.01$. The five with $p < 0.05$ (but > 0.01) are: Wikidata-50, Wikidata-100, Freebase-50, Freebase-75, Freebase-100.
- **Transductive:** All 3 improvements significant at $p < 0.01$.

Stability across runs. The maximum standard deviation observed across all 27 configurations and 3 metrics is 0.014 for MRR and 1.8% for Hits metrics, confirming high stability. Variance is slightly higher on: (1) harder splits (V3/V4) where test graphs are sparser, (2) low-resource settings (25% relation overlap), and (3) noisy datasets (NELL-995). This is expected as these settings have higher intrinsic uncertainty.

Table 11: Mean $\pm$ standard deviation over 3 runs for entity-inductive benchmarks (KGWN-L).

Dataset	**MRR**			**Hits@1 (%)**			**Hits@10 (%)**		
	V1/V2	V3	V4	V1/V2	V3	V4	V1/V2	V3	V4
WN18RR									
V1	$.761 \pm .008$	–	–	72.4 ± 1.2	–	–	86.4 ± 0.8	–	–
V2	$.732 \pm .009$	–	–	68.6 ± 1.4	–	–	85.1 ± 0.9	–	–
V3	–	$.491 \pm .011$	–	–	44.8 ± 1.8	–	–	62.3 ± 1.5	–
V4	–	–	$.678 \pm .007$	–	–	64.2 ± 1.1	–	–	76.8 ± 0.9
FB15k-237									
V1	$.502 \pm .013$	–	–	42.5 ± 1.6	–	–	64.5 ± 1.3	–	–
V2	$.556 \pm .012$	–	–	46.8 ± 1.5	–	–	72.4 ± 1.1	–	–
V3	–	$.511 \pm .014$	–	–	42.1 ± 1.7	–	–	68.9 ± 1.4	–
V4	–	–	$.489 \pm .012$	–	–	39.5 ± 1.6	–	–	67.8 ± 1.2
NELL-995									
V1	$.831 \pm .006$	–	–	77.3 ± 0.9	–	–	95.2 ± 0.5	–	–
V2	$.542 \pm .010$	–	–	44.8 ± 1.4	–	–	73.2 ± 1.1	–	–
V3	–	$.528 \pm .011$	–	–	44.6 ± 1.5	–	–	70.5 ± 1.3	–
V4	–	–	$.447 \pm .013$	–	–	36.5 ± 1.7	–	–	59.8 ± 1.5

Table 12: Mean $\pm$ standard deviation over 3 runs for relation-inductive benchmarks (KGWN-L).

Metric	**NELL**				**Wikidata68K**				**Freebase**			
	100%	75%	50%	25%	100%	75%	50%	25%	100%	75%	50%	25%
MRR												
Mean	.342	.291	.312	.371	.128	.278	.091	.215	.251	.218	.148	.168
Std	.009	.011	.010	.008	.012	.010	.014	.011	.011	.012	.013	.012
Hits@1 (%)												
Mean	24.1	19.6	22.3	27.8	9.5	20.8	5.8	15.4	17.6	14.8	9.6	10.1
Std	1.3	1.5	1.4	1.2	1.6	1.4	1.8	1.5	1.5	1.6	1.7	1.6
Hits@10 (%)												
Mean	54.8	50.6	49.5	54.7	20.1	39.8	16.8	34.8	41.2	36.8	26.1	31.5
Std	1.1	1.3	1.2	1.0	1.5	1.2	1.6	1.3	1.2	1.4	1.5	1.4

Table 13: Mean $\pm$ standard deviation over 3 runs for transductive benchmarks (KGWN-L).

Dataset	**MRR**	**Hits@1 (%)**	**Hits@10 (%)**
WN18RR	$.598 \pm .007$	54.8 ± 1.0	69.8 ± 0.8
FB15k-237	$.448 \pm .010$	35.2 ± 1.3	63.4 ± 1.0
NELL-995	$.589 \pm .008$	52.8 ± 1.1	69.6 ± 0.9